FPGA-based Neuromorphic Acceleration of Spiking Neural Networks

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1. Summary of the Research

Motivation

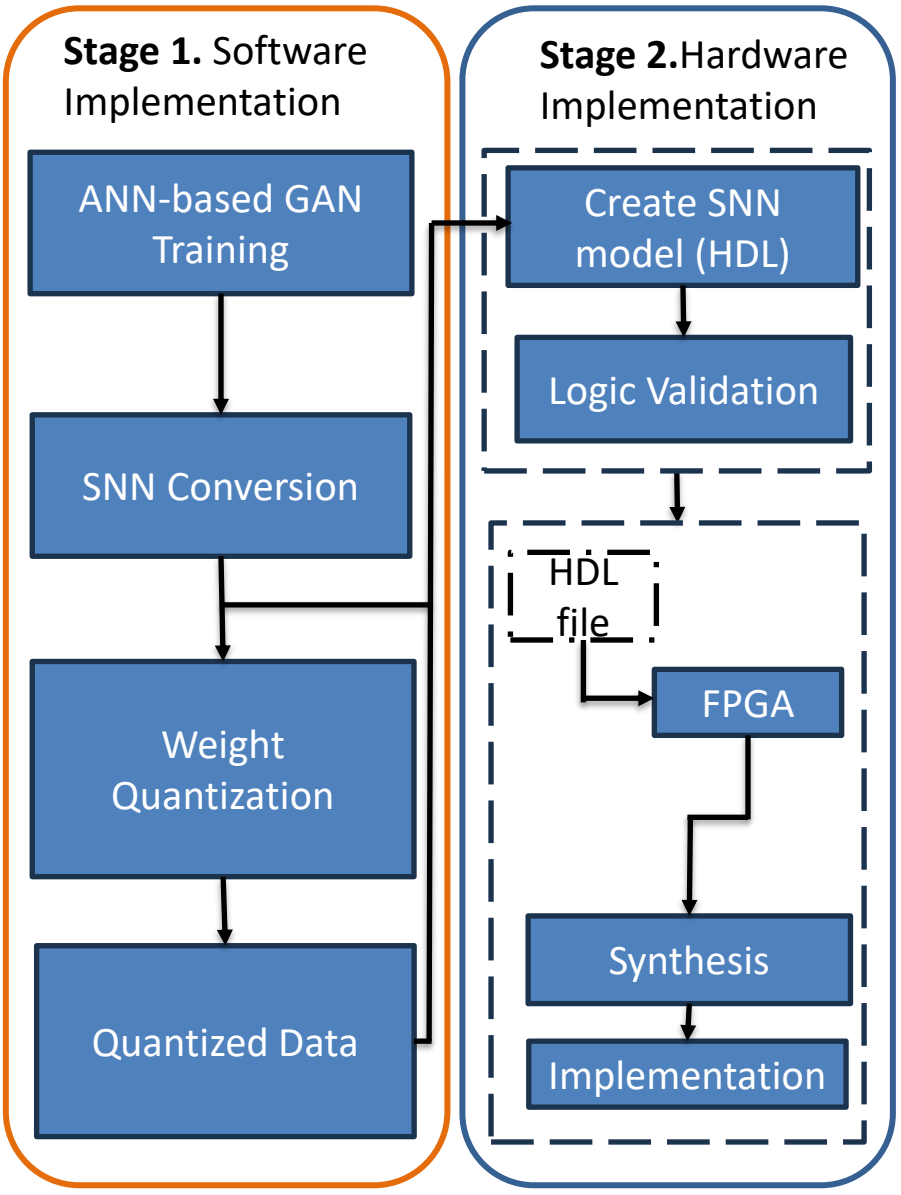
Generative Adversarial Networks (GANs)[1] are a foundational technology in generative AI, achieving strong performance in content generation. But their high computational cost limits deployment on devices, which have strict power budgets. We attempt to solve this by utilizing event-driven, multiplication-free computation based on Spiking Neural Networks (SNNs)[2].

Goal

Implementation of a Low-Power Spiking Neural Network (SNN)-Based GAN Generator on the AMD VCK5000 Versal FPGA Platform

2. Approach/Methodology

The above figure is the design flow of our proposed framework.



In **Stage 1**, an ANN-based GAN is trained and converted into an SNN. Critical network parameters, such as weights, biases, and input spikes, are extracted and quantized.
In **Stage 2**, the quantized parameters in Stage 1 are integrated into on-chip memory modules. RTL simulation is performed for logic validation, followed by FPGA synthesis and device implementation.

3. Current Results and Status

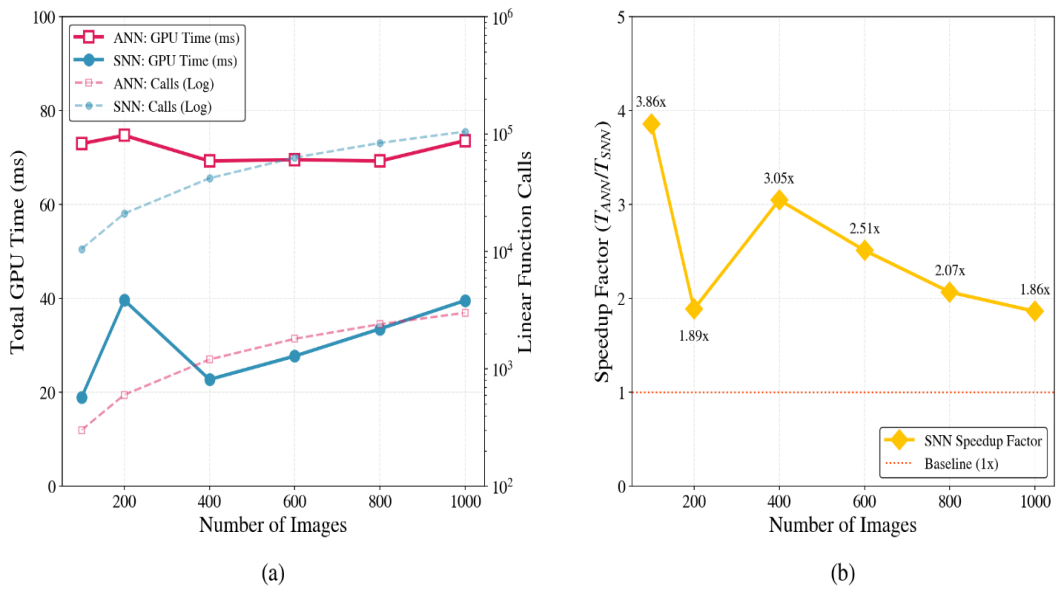


Figure 3. Performance analysis of ANN and SNN generators:
a) Total GPU execution time and linear function calls across different image scales.
b) SNN speedup ratio relative to the ANN.

4. Remaining Tasks and Tentative Schedule

	10	11	12	1
[a]	→	→		
[b]	→	→	→	→

[a] Implementation on FPGA

[b] Writing a thesis

References

[1] Goodfellow, Ian, et al. "Generative adversarial networks." Communications of the ACM 63.11 (2020): 139-144.
[2] Merolla, Paul A., et al. "A million spiking-neuron integrated circuit with a scalable communication network and interface." Science 345.6197 (2014): 668-673